Radiologist Adjudication and Inter-Rater Agreement for Automated Radiology Report Scoring

A Study Protocol

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Abstract

LAIBench scores finding-text-to-report generation with deterministic clinical checks, a hard critical-finding gate, and an optional frozen judge. To date its labels are an internal data-quality process, not an independent third-party validation. This protocol pre-registers a blinded multi-reader study to estimate (i) inter-rater agreement among board-certified radiologists on the report-faithfulness constructs LAIBench measures, and (ii) the agreement between the LAIBench score and a radiologist consensus. We define the locked subset, blinding, disagreement-resolution procedure, agreement statistics, and acceptance thresholds in advance. **No results are reported here**; this document fixes the design before any data is collected.

1 Objectives

1. Estimate inter-rater reliability (Cohen/Fleiss κ, Krippendorff α) among radiologists for: critical-finding preservation, clinically relevant omission, hallucinated/unsupported finding, contradiction, and impression actionability.
2. Estimate scorer–human agreement: correlation and rank concordance between the LAIBench per-case score (and its CRIT gate) and the radiologist consensus.
3. Identify systematic scorer failure modes (where the deterministic checks and the human panel disagree) to drive calibration.

2 Design

Sample. A *locked* subset of N report instances drawn from the controlled-eval suite, frozen by suite hash before reading begins. Cases stratified by modality, anatomy, and presence/absence of a critical finding (over-sampling critical and negative-control cases).

Readers. 2–3 board-certified radiologists, plus at least one reviewer *not affiliated with Laudos.AI*. Readers are blinded to the system identity, to each other's ratings, and to the LAIBench score.

Rating instrument. Per case, each reader records a structured judgment on the constructs in Objective 1, with a free-text justification, on a fixed rubric agreed before reading.

Disagreement protocol. Discordant cases are resolved by a pre-specified rule (independent third reader, then consensus discussion); the adjudicated label and the raw per-reader labels are both retained.

3 Analysis plan

Inter-rater: report κ/α with bootstrap confidence intervals per construct. Scorer–human: Spearman ρ and Pearson r between LAIBench score and consensus, plus a confusion analysis of the binary critical-finding gate against the human critical-finding judgment (sensitivity/specificity of the gate). Calibration: reliability diagram of scorer score versus probability of human "acceptable". Pre-specified acceptance threshold for "scorer tracks radiologists": consensus correlation and gate sensitivity above thresholds fixed in the locked protocol.

4 Governance and limitations

The subset is locked and never used to tune the scorer (that would destroy it as a holdout). This is a measurement study of agreement, not a clinical trial: it does not establish clinical safety, diagnostic accuracy, or regulatory fitness, and it is not image interpretation. Until this study reports, the correct public claim for LAIBench remains *technical benchmark framework*, not *clinically validated benchmark*. Reader counts, exact N, and thresholds are finalized in the locked protocol prior to data collection. Validator and record schema for adjudication artifacts already exist in the LAIBench repository; this protocol governs the study that produces them.